

Math 140 Spring 2023 - Quiz 6

Section:

Name:

Show all your work.

1. Use the First Derivative Test or the Second Derivative Test to determine the relative extreme values of the function $f(x) = 2x^3 + 12x^2 + 78$.

See other page.

2. A rectangular box open at the top has vertical sides, a "square" bottom, and a volume of 32 cubic meters. Find its dimensions having the least possible surface area. (Hint: First, write the surface area as a function of one variable only.)

See other page.